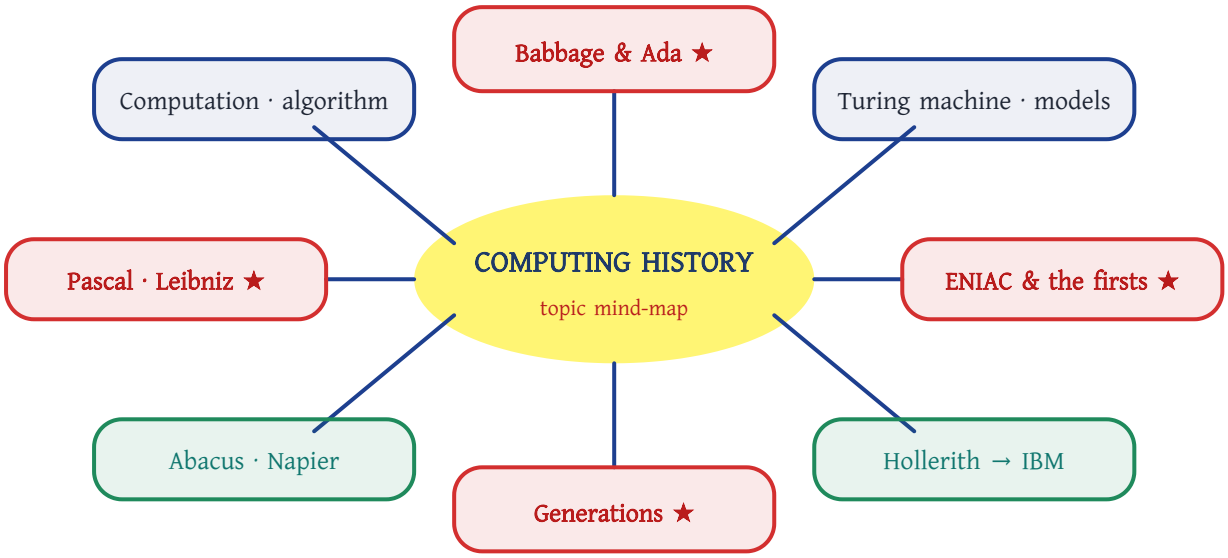
Computation & History of Computing

Algorithms · models · abacus to ENIAC · stored program · generations: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = supporting facts

Syllabus coverage

"Introduction to Computer Systems: **Introduction to computations**; **Early history of computing devices**; Computers; Major components of a computer ...". This notebook covers computation (algorithms, types, models) and the history from the abacus to the computer generations; computer types, components and architecture are in the next lessons of this chapter.

High-yield list

① father of computer: Babbage
② first programmer: Ada
③ Pascaline vs Leibniz
④ ENIAC: 17,468 tubes
⑤ von Neumann: stored program
⑥ generation ↔ technology

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

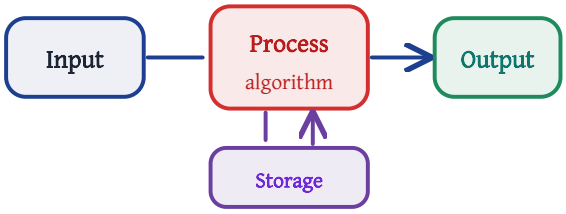
- ✓ Computation & algorithms ..... p. 2
- ✓ Models of computation ..... p. 3
- ✓ Abacus to mechanical calculators ... p. 4
- ✓ Babbage & Ada Lovelace ..... p. 5
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1 What Is Computation?

Definition: **Computation** is the transformation of **input into output** by a finite sequence of well-defined steps, i.e. by an **algorithm**. It can be carried out by a person, a mechanical device or an electronic computer. Until the 1940s "computer" was a **job title** for a person who did calculations.

Input → Algorithm → Output

- **Input:** the data given (e.g. two numbers, a list of names)
- **Process:** the rules / algorithm applied
- **Output:** the result (sum, sorted list)
- **Storage** keeps data between steps; this is the IPO(S) cycle of every computer



the IPO model with storage

Algorithm: the five properties (Knuth)

Property	Meaning
Finiteness	must stop after a finite number of steps
Definiteness	every step is precise and unambiguous
Input	zero or more inputs
Output	**at least one** output
Effectiveness	each step is basic enough to be done exactly, by pencil and paper

Memory trick: **F**IDO-E: **F**initeness, **I**nput, **D**efiniteness, **O**utput, **E**ffectiveness. "FIDO the dog is Effective."

Word origin: "Algorithm" comes from **al-Khwarizmi** (c. 780–850, Baghdad), whose book *al-jabr* also gave us "algebra".

Worked example: Euclid's algorithm (c. 300 BCE) ✍️: one of the oldest algorithms still in use. gcd(252, 105):

1. 252 = 2 × 105 + **42** → gcd(105, 42)
2. 105 = 2 × 42 + **21** → gcd(42, 21)
3. 42 = 2 × 21 + **0** → stop: gcd = **21** (finite, definite, one output ✓)

Exam trap: A **program** is an algorithm written in a programming language; a procedure that may run forever (e.g. an OS loop) is not an algorithm in the strict sense, because it lacks **finiteness**.

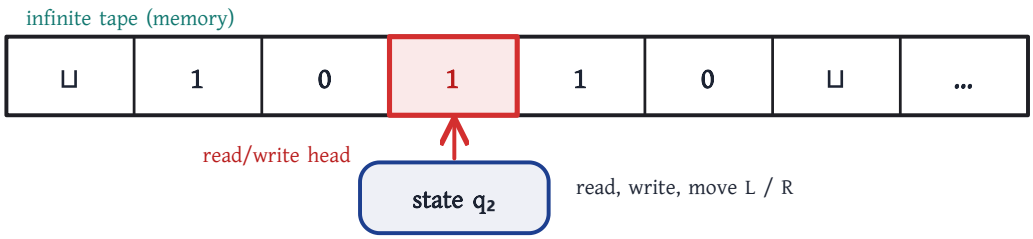
2 Types of Computation

Type	What it does	Example
Exact	precise, deterministic result	5 × 7 = 35; integer arithmetic
Approximate	result close to the true value	√2 ≈ 1.41421 (Newton's method)
Symbolic	manipulates symbols, not numbers	d/dx (x^2) = 2x; Mathematica
Logical / Boolean	true / false decisions	if (marks ≥ 40 && present)

Exam trap: Digital computers also compute **approximately**: in floating point 0.1 + 0.2 = 0.30000000000000004, because 0.1 has no exact binary form (see Number Systems, cs5).

3 Models of Computation

Definition: A **model of computation** is an abstract machine that defines **what can be computed** and what each step costs.



Turing machine: a finite control + a head + an unbounded tape

Model	Proposed by	Memory	Power / use
Finite automaton (FA)	Kleene, 1956	current state only	regex, lexers, vending machines
Pushdown automaton	1960s	one stack	compiler parsers
Turing machine	**Turing, 1936**	unbounded tape	everything computable
Lambda (λ) calculus	Church, 1936	functions only	= TM; basis of Lisp, Haskell

Key ideas:

Church–Turing thesis: whatever is effectively computable is computable by a **Turing machine**. **Universal TM:** one machine runs any other's description (→ stored program). **Halting problem: undecidable.**

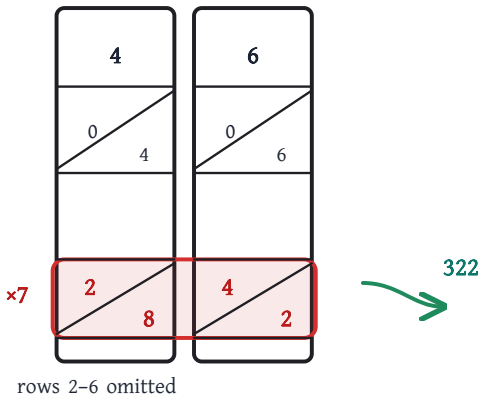
Memory trick: Power ladder: **State < Stack < Tape** (FA < PDA < TM).

Exam trap: An FA cannot check balanced brackets ($a^n b^n$): it has no memory beyond its state. A PDA can.

4 Counting Aids & Mechanical Calculators

Device	Year	Inventor	Remember
Tally sticks · quipu	ancient	notches · Inca knots	recording, not calculating
Abacus	c. 2400 BCE	Mesopotamia	**first calculating device**
Antikythera mechanism	c. 100 BCE	Greece	oldest known **analog** computer
Napier's bones	1617	John Napier	rods for ×; **logarithms** 1614
Slide rule	c. 1622	William Oughtred	analog, log scales
Calculating clock	1623	Wilhelm Schickard	earliest mechanical calculator
Pascaline	**1642**	Blaise Pascal	gears: **add & subtract only**
Stepped Reckoner	1673	G. W. Leibniz	**all four operations**
Jacquard loom	1804	J.-M. Jacquard	**punched cards** set the pattern
Arithmometer	1820	Thomas de Colmar	first mass-produced calculator

Napier was Scottish, Oughtred English, Schickard and Leibniz German, Pascal, Jacquard and de Colmar French. The abacus survives as the Chinese *suanpan* and Japanese *soroban*. Pascal built the Pascaline at 19 to help his father, a tax official; the Antikythera gears (found in a shipwreck in 1901) predicted planet positions and eclipses.



Napier's bones: 46 × 7

Worked example ✍️: 46 × 7 with Napier's bones: take the rods for 4 and 6, read row 7 → **2/8** and **4/2**. Add along the diagonals from the right: units = 2; tens = 8 + 4 = 12 → write 2, carry 1; hundreds = 2 + 1 = 3 → **322** ✓

Memory trick: **1642**: the Pascaline appeared in the year Galileo died and Newton was born. Napier → Schickard → Pascal → Leibniz = **17, 23, 42, 73** (all 16xx).

Exam trap: The Pascaline could only **add and subtract** (it subtracted with nines' complements). Leibniz's machine was the first to do all **four** operations. If an MCQ asks for the "first mechanical calculator" without listing Schickard, answer **Pascaline**.

IT link: The programming language **Pascal** (Wirth, 1970) is named after Blaise Pascal; Leibniz also described the **binary number system** (1703), which every digital computer uses.

5 Charles Babbage: the Engines

Who: **Charles Babbage** (1791–1871), English mathematician: the **father of the computer**, because his Analytical Engine had every part of a modern computer.

Difference Engine:
Proposed **1822**. Tabulates polynomials by the **method of finite differences**: **only addition**. Special-purpose; unfinished. No. 2 was built by London's Science Museum in **1991**: it works.

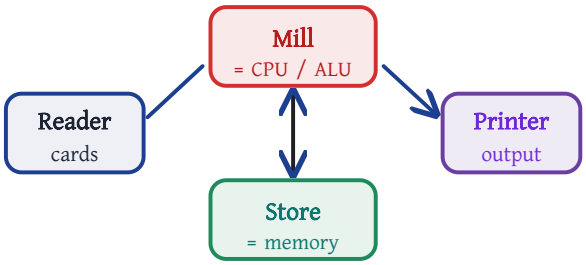
Analytical Engine:
Designed **1834–37**. **General-purpose, programmable**, steam-powered; punched cards (from Jacquard), loops, **conditional branching**; store of 1,000 40-digit numbers. Never built.

Worked example: method of differences ✍️: $f(x) = x^2 + x + 41$

x	0	1	2	3	4	5
f(x)	41	43	47	53	61	71
Δ^1	2	4	6	8	10	
Δ^2	2	2	2	2		

Degree 2 → Δ^2 is constant. Next $\Delta^1 = 8 + 2 = 10$; next $f = 61 + 10 = 71$ (check: $25 + 5 + 41 = 71$). Only two additions, no multiplication.

Analytical Engine → modern computer



Babbage's four parts (control by a "barrel")

Analytical Engine	Today
Reader (punched cards)	input unit
Mill	**CPU / ALU**
Store	**memory**
Barrel (pegged drum)	control unit
Printer / curve plotter	output unit

6 Ada Lovelace: the First Programmer

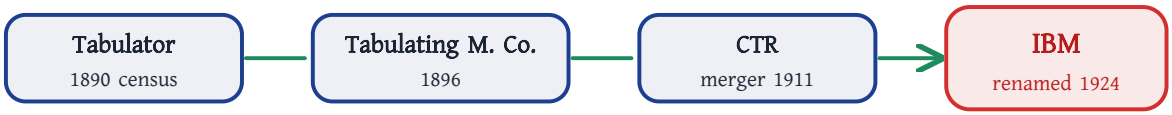
Who: **Ada Lovelace** (1815–1852), daughter of Lord Byron. In **1843** she translated Menabrea's paper on the engine and added Notes A–G; **Note G**, an algorithm for **Bernoulli numbers**, is the first published program. She foresaw computing with symbols and music.

Memory trick: The **Mill** "mills" (processes) numbers; the **Store** stores them. The language **Ada** (US DoD, 1980) is MIL-STD-**1815**: her birth year.

Exam trap: Babbage **completed neither** engine. The Analytical Engine is the first **concept** of a programmable computer, not a working one.

7 Punched Cards: Hollerith to IBM

Herman Hollerith: built an electric **tabulating machine** that read punched cards for the **1890 US Census**. The 1880 census took about 8 years to process; the 1890 count took about one year. This was the first large-scale **data processing** by machine.



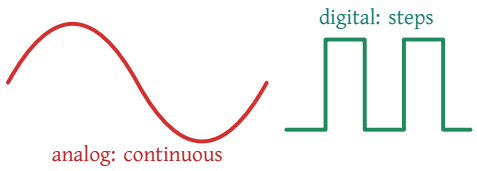
- **Comptometer** (Dorr Felt, 1887): first commercially successful **key-driven** calculator.
- IBM's **80-column punched card** (1928) became the standard; old terminals showed 80 characters per line for this reason.

8 Analog, Digital & Hybrid

	Analog	Digital	Hybrid
Works on	**continuous** quantities	**discrete** digits (0 / 1)	both
Accuracy	limited by measurement	**high** (set by bits)	in between
Examples	slide rule, **differential analyzer**	abacus, ENIAC, PCs	ICU monitor, petrol pump

The **differential analyzer** (Vannevar Bush, MIT, **1931**) solved differential equations with rotating shafts: the most powerful analog computer before ENIAC.

Exam trap: The **abacus is digital** (beads are discrete counts) while the **slide rule is analog** (a continuous scale). A thermometer and a car speedometer with a needle are analog.



9 The Theory Arrives (1854–1948)

Year	Who	Contribution
1854	George Boole	**Boolean algebra** (*The Laws of Thought*)
1936	Alan Turing	Turing machine: the limits of computation
1937	Claude Shannon	Boolean algebra describes **switching circuits**
1948	Claude Shannon	information theory; the word **"bit"** in print

Memory trick: **Boole thinks (1854), Turing defines (1936), Shannon wires (1937)**: logic → theory → circuits.

Exam trap: The word "bit" (binary digit) was **coined by John Tukey**; Shannon first used it in print (1948).

10 Relays to Vacuum Tubes: the First Computers

Machine	Year	Built by	Technology	The first ...
Z3	1941	Konrad Zuse	relays, binary	working **programmable**
ABC	1942	Atanasoff–Berry	~300 tubes, binary	**electronic digital**
Colossus	1944	Tommy Flowers	1,600+ valves	**programmable electronic**
Harvard Mark I	1944	Aiken + IBM	relays, decimal	big electromechanical
ENIAC	**1946**	Eckert–Mauchly	tubes, **decimal**	**general-purpose electronic**

- **Z3** (Berlin, May 1941, ~2,600 relays) used binary floating point and read its program from punched film; destroyed in a 1943 air raid.
- **Colossus** (Bletchley Park) broke the German **Lorenz** teleprinter cipher; it stayed secret until the 1970s.
- **Harvard Mark I** = IBM ASCC: about 15 m long, read instructions from paper tape; Grace Hopper was one of its programmers.

ENIAC = decimal, Z3 & ABC = binary!

ENIAC by the numbers

17,468	~30 t	~167 m²	150 kW	5,000
vacuum tubes	weight	1,800 sq ft floor	power	additions / s

Electronic Numerical Integrator and Computer · built for US Army artillery firing tables · unveiled Feb 1946

Programming ENIAC: by setting switches and **rewiring plugboards**: a new problem took days. Its six first programmers were women. This weakness led directly to the **stored-program** idea (next sheet).

Exam traps: ① ENIAC worked in **decimal**. ② Colossus broke **Lorenz**; **Enigma** was attacked with Turing's electromechanical **Bombe**. ③ ABC was **not programmable**; a 1973 US court voided the ENIAC patent, crediting Atanasoff.

Memory trick: Alphabet = time order: **Z**use (Z3, '41) → **A**BC ('42) → **C**olossus ('44) → **E**NIAC ('46). "Z-ACE".

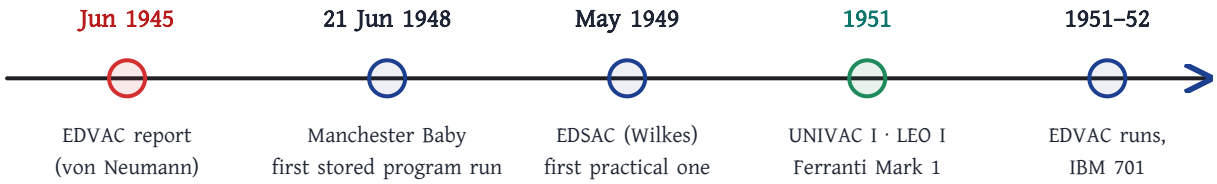
IT link: The first computer "**bug**": a moth stuck in a relay of the Harvard **Mark II** (9 Sept 1947), taped into the log book; Grace Hopper made "debugging" popular.

11 The Stored-Program Idea

Definition: In a **stored-program computer** the instructions are kept in the **same read-write memory** as the data, so a new program is loaded, not wired. Described by **John von Neumann** in the *First Draft of a Report on the EDVAC* (June **1945**), drawing on Eckert, Mauchly and Turing's universal machine.

- binary representation
- one memory for program + data
- instructions fetched and executed **one after another** (fetch–decode–execute)
- limitation: the **von Neumann bottleneck** (one path to memory)

Next lesson: CPU, buses, von Neumann vs **Harvard** (separate program and data memory, as in the Mark I) → **cs2**, Hardware & Architecture.



the stored-program timeline

- **UNIVAC I** (Eckert–Mauchly, 1951): first **commercial computer in the USA**; first unit to the **US Census Bureau**; magnetic tape; predicted the 1952 election on TV.
- **Ferranti Mark 1** (UK, Feb 1951) came weeks earlier; **LEO I** (1951) ran the first business job.

12 The "Firsts" Table

"The first ..."	Answer
calculating device	abacus
mechanical calculator (usual MCQ answer)	Pascaline, 1642 (Schickard 1623 if listed)
programmable computer (concept)	Analytical Engine, Babbage
working programmable digital computer	Z3, 1941
electronic digital computer	ABC, 1942
general-purpose electronic computer	**ENIAC**
stored program to run	Manchester Baby, 1948
commercial computer (USA)	UNIVAC I, 1951

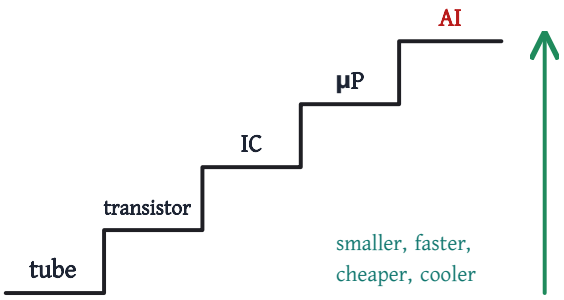
Exam trap: ENIAC was **not** stored-program as built; EDVAC's report proposed it, the **Baby** ran it first. Read "first" questions word by word.

Memory trick: **B-E-E**: Baby (1948) → EDSAC (1949) → EDVAC (1951). The report came first, the machine last.

13 Generations of Computers

Basis: A generation is named after its **main switching technology**. Dates differ by a few years between books, so **answer by technology, not by year**.

Gen	Period	Technology	Memory · software	Examples
1st	1946–59	**vacuum tubes**	drum; machine language	ENIAC, UNIVAC I
2nd	1959–65	**transistors**	core; assembly, FORTRAN	IBM 1401, IBM 1620
3rd	1965–71	**ICs** (SSI/MSI)	OS, multiprogramming	System/360, PDP-8
4th	1971–now	**microprocessor**	GUI, networks	IBM PC, Apple II
5th	future	ULSI, parallel, **AI**	natural language	Japan's FGCS (1982)



each step up: size ↓ speed ↑ cost ↓

Memory trick: **V-T-I-M-A**: "**V**ery **T**ired **I**nterns **M**ake **A**pps" = Vacuum tube, Transistor, IC, Microprocessor, AI.

Exam trap: IBM System/360 = **3rd** generation; IBM 1620 and 1401 = **2nd**; "VLSI / microprocessor" = **4th**; "AI, natural language" = **5th**.

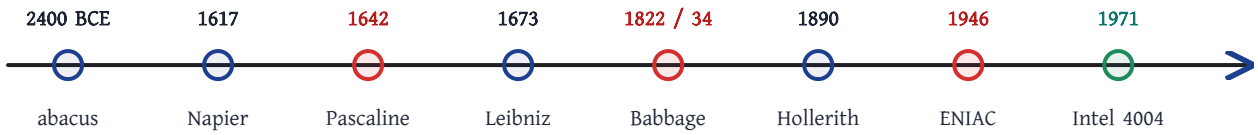
Inventions behind each step

Invention	Year	By / fact
Transistor	1947	Bardeen, Brattain, Shockley (Bell Labs)
Integrated circuit	1958	**Jack Kilby** (TI); Robert Noyce, silicon IC 1959
Microprocessor **Intel 4004**	**1971**	Hoff, Faggin, Mazor: **4-bit**, 2,300 transistors
Moore's law	1965	transistors per chip double ~every 2 years
Personal computer	1975 / 81	Altair 8800 (Ed Roberts) / IBM PC

Bangladesh: The first computer in the country was an **IBM 1620** (2nd generation, transistorised), installed in **1964** at the **Atomic Energy Centre, Dhaka**. Bangladesh Computer Council (BCC) was set up by an Act of 1990.

Characteristics of computers and types by size (super, mainframe, mini, micro) → **cs2**; system and application software → **cs3**.

Night-Before-the-Exam Sheet



eight dates carry most of the marks

Who's who: the "fathers"

Title	Person	Title	Person
father of the computer	**Charles Babbage**	first programmer	**Ada Lovelace**
father of modern computing / AI theory	Alan Turing	stored-program architecture	John von Neumann
term "artificial intelligence" (1956)	John McCarthy	father of the personal computer	Ed Roberts (Altair)
first compiler (A-0, 1952), COBOL	Grace Hopper	FORTRAN (1957)	John Backus
information theory	Claude Shannon	supercomputer (CDC 6600)	Seymour Cray

Top 12 one-liners for MCQs

1. Algorithm ← al-Khwarizmi
2. Turing machine: 1936
3. Napier's bones: 1617
4. Pascaline: add & subtract only
5. Leibniz: all four operations
6. Punched cards: Jacquard loom 1804
7. Analytical Engine: Mill + Store
8. Hollerith → IBM (1924)
9. ENIAC: 17,468 tubes, decimal
10. Colossus: Lorenz cipher
11. UNIVAC I: first US commercial
12. BD's first: IBM 1620, 1964

Top traps: Schickard before Pascal · Pascaline ≠ four operations · Babbage finished neither engine · ABC not programmable · ENIAC decimal and not stored-program · Colossus ≠ Enigma · Baby ran the first stored program, EDVAC only proposed it · "bit" coined by Tukey · answer generation questions by technology

All tricks together: FIDO-E · State < Stack < Tape · 1642 = Galileo dies, Newton born · Mill = CPU, Store = memory · Ada = MIL-STD-1815 · Boole thinks, Turing defines, Shannon wires · Z-ACE · B-E-E · V-T-I-M-A

✓ Practice MCQs (exam pattern, 10 marks)

1. The Pascaline (1642) could:
   (a) × ÷ + − (b) + and − only
   (c) × only (d) logarithms
2. Who is called the father of the computer?
   (a) von Neumann (b) Alan Turing
   (c) Blaise Pascal (d) Charles Babbage
3. First-generation computers used:
   (a) vacuum tubes (b) transistors
   (c) ICs (d) microprocessors
4. Colossus was built to break the:
   (a) Enigma (b) Purple code
   (c) Lorenz cipher (d) Caesar cipher

5. ENIAC did its arithmetic in:

(a) binary (b) octal
(c) hexadecimal (d) decimal

6. The first stored program was run on:

(a) ENIAC (b) the Baby
(c) UNIVAC I (d) Mark I

7. A Turing machine has, but an FA lacks:

(a) states (b) an alphabet
(c) a tape (d) a start state

8. "Each step precise and unambiguous" is:

(a) definiteness (b) finiteness
(c) effectiveness (d) output

9. The Intel 4004 (1971) was a:

(a) 8-bit CPU (b) 4-bit CPU
(c) 16-bit CPU (d) 32-bit CPU

10. Babbage's "Store" is today's:

(a) memory (b) ALU
(c) input (d) output

Answer key → 1-b · 2-d · 3-a · 4-c · 5-d · 6-b · 7-c · 8-a · 9-b · 10-a

Written question: Describe Babbage's Analytical Engine, relate its parts to a modern computer, and explain why Ada Lovelace is called the first programmer. 8 marks

1. **What:** designed 1834–37 by Charles Babbage; a general-purpose, programmable, steam-powered mechanical computer; never completed. 1
2. **Parts:** Reader (punched cards, from Jacquard) = input; **Mill** = CPU/ALU; **Store** (1,000 × 40-digit numbers) = memory; barrel = control unit; printer = output. 3
3. **Why "computer":** separate processor and memory, program on cards, loops and **conditional branching**: the logical design of a modern computer. 2
4. **Ada Lovelace:** her 1843 Note G gave an algorithm for Bernoulli numbers on the engine, the first published program; she foresaw non-numeric computing. 2

Before the exam ✓

- ✓ 5 properties of an algorithm
- ✓ abacus → … → ENIAC with years
- ✓ every "first" and every "father"
- ✓ generation ↔ technology

Also practise (written):

① Analog vs digital vs hybrid
② Generations with examples
③ Turing machine and its significance
④ Method of differences for $x^2 + 1$